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AI/ML APPLICATION FOR WEATHER DATA PREPROCESSING, VISUALIZATION, AND EXTREME RAINFALL PREDICTION.

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Introduction

In weather forecasting, the predicting atmospheric conditions is vital for sectors such as agriculture, tourism, airport operations, mining, and power generation. Managing data from automatic weather station systems requires meticulous data cleaning to address challenges like missing data and outliers in continuous streams. Ensuring data integrity and reliability relies on specialized data preprocessing techniques. The transition of weather forecasting into Big Data, driven by advancements in climate observing systems and increased data volume, deems traditional computational models inadequate and prompts the need to address gaps in prediction accuracy. Consequently, the adoption of advanced AI/ML-based data preprocessing becomes imperative for processing extensive datasets, with a focus on utilizing deep learning techniques to achieve accurate predictions.

Objectives

1. Acquire understanding of predicting atmospheric conditions at specific locations by considering multivariable analysis of various weather parameters.
2. Explore the integration of weather forecasting within the domains of data preprocessing and Big Data analytics, recognizing advancements in Artificial Intelligence and Machine Learning (AIML), meteorological observation technologies, and the notable expansion in weather data volume.
3. Investigate the move towards employing deep learning techniques to process extensive datasets for weather forecasting, elevating the learning process and facilitating more precise predictions.

Methodology & Study Area

Machine Learning offers effective algorithms for analyzing and forecasting rainfall patterns. This research explores the use of **LSTM (Long Short-Term Memory)** models for weather prediction based on time series weather data. The **India Meteorological Department (IMD)** has established a comprehensive network of dense surface observations, which incorporate state-of-the-art Automatic Weather Stations (**AWS**) across the country. This study utilized IMD AWS data collected from an Automatic Weather Station situated in **Uttarakhand's Almora district**, spanning from January 2020 to September 2023.

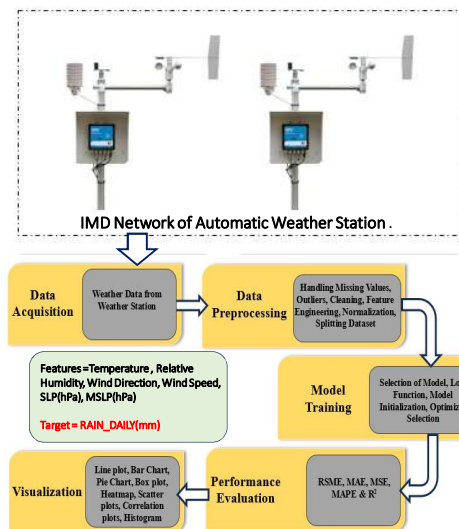


Fig. 1. Proposed weather forecasting model.[3]

Experimental Setup & Results

The dataset is from the IMD AWS station in Almora, Uttarakhand, with coordinates approximately 29.6002° N latitude and 79.6651° E longitude. The elevation is 1620.0 meters, and the data covers January 2020 to September 2023.

Table I. Meteorological Attributes

ATTRIBUTES	NON-NULL	MISSING VALUES	% MISSING
DATE/TIME	32868	0	0
RAIN_DAILY(mm)	28591	4277	13.01
TEMP(C)	25831	4337	13.20
RH(%)	28601	4267	12.98
WIND_DIR_10M(DEG)	28602	4266	12.98
WIND_SPEED_10M(KT)	28601	4267	12.98
SLP(hPa)	28531	4337	13.20

The dataset, containing approximately 13% missing values, was mitigated using spline interpolation. Additionally, 1% of outliers were addressed through regression technique. The dataset is divided, allocating 80% of the data for training the model and reserving the remaining 20%.

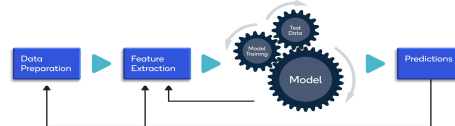


Fig. 2. AI/ML development pipeline[2]

In this research, a deep learning architecture based on recurrent neural network, specifically the LSTM model, was employed for rainfall prediction, exploring various parameters. The LSTM cell is designed to capture and learn long-term dependencies in sequential data.

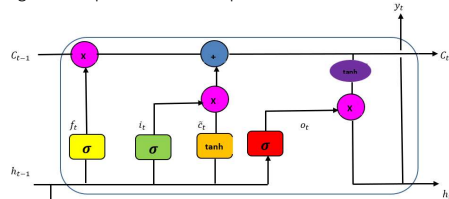


Fig. 3. Block diagram of LSTM model[3]

$$y_t = \text{Activation}(W_{\text{output}} * h_t + b_{\text{output}}) \quad (1)$$

TABLE II. LSTM Model Configurations

MODEL	LAYERS	ACTIVATIONS	OPTIMIZERS	EPOCHS	LEARNING RATE	BATCH SIZE
MODEL-1	1	relu	adam	100	0.000001	32
MODEL-2	1	relu	adam	100	0.001	72
MODEL-3	1	relu	adam	300	0.1	72
MODEL-4	2	relu	adam	100	0.000001	72
MODEL-5	3	relu	adam	100	0.000001	72
MODEL-6	3	relu	adam	300	0.5	200
MODEL-7	5	relu	adam	100	0.000001	32

Models can be assessed for their performance using metrics such as RMSE, MSE, MAE, and R².

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (2)$$

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (3)$$

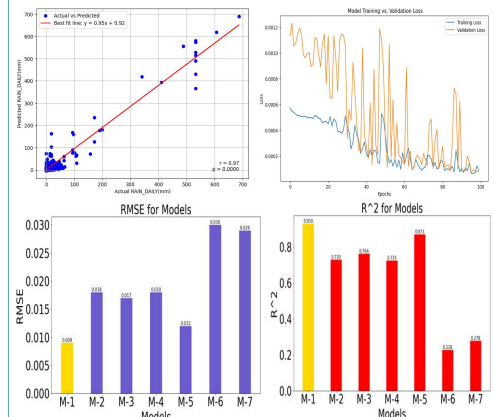
$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (4)$$

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (5)$$

TABLE III. Model Performance Metrics

MODEL	RMSE	MSE	MAE	R ²
MODEL-1	0.009	0.0001	0.004	0.930
MODEL-2	0.018	0.0003	0.004	0.730
MODEL-3	0.017	0.0003	0.006	0.764
MODEL-4	0.018	0.0003	0.005	0.725
MODEL-5	0.012	0.0002	0.005	0.871
MODEL-6	0.030	0.0009	0.005	0.228
MODEL-7	0.029	0.0009	0.004	0.278
(R. K. Grace et al. 2020) LR	3.943	13.280	-	-
(R. K. Grace et al. 2020) MLR	3.449	11.894	-	-

Model-1 have the lowest values for RMSE, MSE, and MAE, seen that it performs well in terms of minimizing errors between predicted and actual values. has the highest R² value, indicating a good fit between the predicted values and the actual values.

Fig. 4. actual vs predicted Rain, Training & Validation loss, Model Performance RMSE & R² vs Models.

Conclusion

- Model-1 fits well with real data, showing minimal errors (RMSE 0.009, MSE 0.0001, MAE 0.004) and a strong R² value of 0.930.
- The accuracy of predictions relies on the quality of AWS data, as well as the proper handling of missing values and outliers.
- It is noticed that changes in hyperparameters can impact the performance of the models.
- Increasing the number of layers does not necessarily lead to performance improvement.
- Further studies are necessary to examine the results using long series dataset, GRU and Bi-LSTM models.

Acknowledgements & References

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